Standard Operating Procedure for Resolving QoS/Congestion Event Failure on Arista Access Node MALL-ACC-80

Node: MALL-ACC-80 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_8667

Description

This SOP is designed to address a critical failure domain on the Arista Access Node MALL-ACC-80, which is located at the Malleswaram Node. The equipment, being a part of the access network, is responsible for providing high-speed connectivity to the end-users. The failure domain in question pertains to QoS (Quality of Service) and congestion events, specifically microbursts, interface tail drops, and QoS queue starvation. These events can lead to packet loss, increased latency, and ultimately, a degradation of the overall network performance. The root cause of this failure domain can be attributed to the inability of the access node to handle the incoming traffic efficiently, which may be due to various factors such as misconfigured QoS policies, inadequate buffer sizes, or hardware-related issues. It is essential to identify the root cause and implement the necessary corrective actions to prevent recurrence.

Pre-requisite Checks

- Ensure that the remote hands are dispatched to the Malleswaram Node to facilitate physical access to the equipment.

- Verify the status of the redundant links to ensure that they are operational and can take over in case of a failure.

- Check the power supply to the node and ensure that it is within the recommended operating range.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Inadequate buffer sizes	Medium
Hardware-related issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy, Expected Output: QoS Policy Configuration ------------------------- , Description: This command displays the current QoS policy configuration, which can help identify any misconfigured policies that may be contributing to the congestion event.

Command: show interface statistics, Expected Output: Interface Statistics ---------------------- , Description: This command provides
```

```
detailed statistics about the interface, including the number of
packets received and transmitted, which can help identify any
indications of congestion.

Command: show buffer statistics, Expected Output: Buffer Statistics
------------------ , Description: This command displays the current
buffer statistics, which can help identify any indications of buffer
overflow or underflow.
```

Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: configure terminal qos policy set priority , Description: This step involves verifying the QoS policy configuration and making any necessary corrections to ensure that the policy is correctly configured to manage traffic flow.

Step 2: Step: Increase buffer sizes, Command: configure terminal interface buffer size , Description: This step involves increasing the buffer sizes on the affected interface to prevent buffer overflow or underflow.

Step 3: Step: Replace hardware components, Command: no interface interface replace, Description: This step involves replacing any faulty hardware components that may be contributing to the failure domain.

Post-Check Verification

```
Command: show qos policy, Expected Output: QoS Policy Configuration
------------------------ , Description: This command is used to
verify that the QoS policy configuration has been corrected and is
functioning as expected.

Command: show interface statistics, Expected Output: Interface
Statistics --------------------- , Description: This command is used
to verify that the interface statistics have returned to normal and
that there are no indications of congestion.

Command: show buffer statistics, Expected Output: Buffer Statistics
------------------ , Description: This command is used to verify that
the buffer statistics have returned to normal and that there are no
indications of buffer overflow or underflow.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: configure terminal qos policy rollback, Description: If the fix fails, this step involves restoring the original QoS policy configuration to ensure that the network returns to its previous state.

- Step: Restore original buffer sizes, Command: configure terminal interface buffer size , Description: If the fix fails, this step involves restoring the original buffer sizes to ensure that the network returns to its previous state.

- Step: Restore original hardware components, Command: no interface interface replace original, Description: If the fix fails, this step involves restoring the original hardware components to ensure that the network returns to its previous state.